

Use the following situation to answer the questions that follow.

A driving school owner believes that Instructor A is more effective than Instructor B at preparing students to pass the state's driver's license exam. An incoming class of 100 students is randomly assigned to two groups, each of size 50. One group is taught by Instructor A; the other is taught by Instructor B. At the end of the course, 30 of Instructor A's students and 22 of Instructor B's students pass the state exam. Let p_A be the proportion of Instructor A's students who pass the exam, and p_B be the proportion of Instructor B's students who pass the exam.

_____ 1) Which of the following statements is FALSE?

- (A) We use the pooled sample proportion because we are assuming the null hypothesis is true.
- (B) We do not have to check that the sample size is less than 10% of the population size because we are performing an experiment.
- (C) Because we are not randomly selecting from the population, we should use the pooled population proportion.
- (D) The null hypothesis for this test would be $H_0: p_A - p_B = 0$
- (E) The alternative hypothesis for this test would be $H_a: p_A - p_B > 0$

_____ 2) What is the z test statistic?

- (A) $\frac{0.6 - 0.44}{\sqrt{\frac{.52(.48)}{100} + \frac{.52(.48)}{100}}}$
- (B) $\frac{0.6 - 0.44}{\sqrt{\frac{.6(.4)}{100} + \frac{.44(.56)}{100}}}$
- (C) $\frac{0.6 - 0.44}{\sqrt{\frac{.6(.4)}{50} + \frac{.44(.56)}{50}}}$
- (D) $\frac{0.6 - 0.44}{\sqrt{\frac{(.52)(.48)}{100}}}$
- (E) $\frac{0.6 - 0.44}{\sqrt{(.52)(.48)\left(\frac{1}{50} + \frac{1}{50}\right)}}$

_____ 3) What is the p-value?

- (A) 1.6012
- (B) 0.0547
- (C) 0.6
- (D) 0.1094
- (E) 0.52

_____ 4) Do these results give convincing evidence at the 5% significance level that Instructor A is more effective?

- (A) Because our p-value is lower than 5%, we reject H_0 . We have convincing evidence that Instructor A is more effective.
- (B) Because our p-value is higher than 5%, we fail to reject H_0 . We do not have convincing evidence that Instructor A is more effective.
- (C) Because our p-value is lower than 5%, we reject H_0 . We have convincing evidence that Instructor B's students do not pass the state exam.
- (D) Because our p-value is higher than 5%, we fail to reject H_0 . We do not have convincing evidence that there is a difference between Instructor A and Instructor B.
- (E) Because our p-value is lower than 5%, we reject H_0 . We have convincing evidence that there is a difference between Instructor A and Instructor B.

_____ 5) Which of the following errors could you have made from your conclusion above?

- (A) Type I; saying that Instructor A was more effective when they were really not.
- (B) Type II; saying that Instructor A was more effective when they were really not.
- (C) Type I; saying that we do not have convincing evidence that Instructor A was more effective when they really were.
- (D) Type II; saying that we do not have convincing evidence that Instructor A was more effective when they really were.
- (E) None of the above errors are correct.

6) Create a 95% confidence interval for the difference in the proportion of students who pass the exam between Instructor A and Instructor B. Do we come to the same conclusion as we did in (4)? Why or why not?